

Logarithm of complex no.

$$z = x + iy$$
$$\log(z) = \log(x + iy)$$
$$= \log(re^{i\theta})$$

$$r = \sqrt{x^2 + y^2}$$
$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$= \log r + i\theta$$
$$= \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1}\left(\frac{y}{x}\right)$$

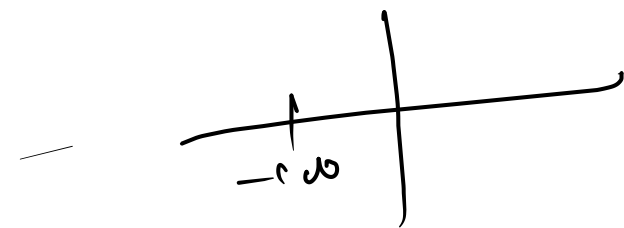
general value

$$\text{Log}(z) = \text{Log}(re^{i\theta})$$

$$= \log(re^{i(2k\pi + \theta)})$$

$$= \log r + i(2k\pi + \theta)$$

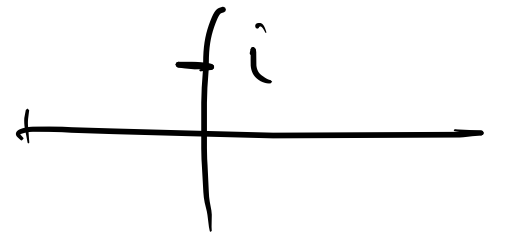
principal value



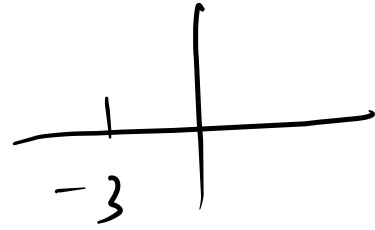
$$\text{Log}(i) = \log(1 e^{i\pi/2}) = \log e^{i\pi/2} = \underline{\underline{i\frac{\pi}{2}}}$$

$$\text{Log}(100) = \text{Log}(100 e^{i\pi}) = \log(100 e^{i(2k\pi + \pi)})$$
$$= \log 100 + i(2k\pi + \pi)$$

$$\begin{aligned}
 \log(i) \\
 &= \log(1 e^{i\pi/2}) \\
 &= i\pi/2
 \end{aligned}$$



$$\begin{aligned}
 \text{Log}(-100) \\
 &= \text{Log}(100 e^{i\pi}) \\
 &= \log(100 e^{i(2k\pi + \pi)}) \\
 &= \log 100 + i(2k\pi + \pi)
 \end{aligned}$$



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- Considering the principal value only prove that $\log_2(-3) = \frac{\log 3 + i\pi}{\log 2}$

$$\begin{aligned}\log_2(-3) &= \frac{\log(-3)}{\log 2} = \frac{\log(3e^{i\pi})}{\log 2} \\ &= \frac{\log 3 + i\pi}{\log 2}\end{aligned}$$

- Find the general value of $\text{Log}(1+i) + \text{Log}(1-i)$

$$\begin{aligned}\text{Log}(1+i) &= \text{Log}(\sqrt{2} e^{i\pi/4}) \\ &= \log(\sqrt{2} e^{i(2k\pi + \pi/4)}) \\ &= \log \sqrt{2} + i(2k\pi + \pi/4)\end{aligned}$$

$$\begin{aligned}\text{Log}(1-i) &= \text{Log}(\sqrt{2} e^{-i\pi/4}) \\ &= \log(\sqrt{2} e^{-i(2m\pi + \pi/4)}) \\ &= \log \sqrt{2} - i(2m\pi + \pi/4)\end{aligned}$$

$$\begin{aligned}\text{Log}(1+i) + \text{Log}(1-i) &= 2 \log \sqrt{2} + i(2k\pi + \pi/4) - i(2m\pi + \pi/4) \\ &= \frac{2}{2} \log 2 + i(2k\pi + \cancel{\pi/4} - 2m\pi - \cancel{\pi/4}) \\ &= \log 2 + i 2\pi(k-m)\end{aligned}$$

Find the value of $\log [\sin(x + iy)]$

$$\cosh^2 y - \sinh^2 y = 1$$

$$\sin(x + iy)$$

$$= \sin x \cos iy + \cos x \sin iy$$

$$= \sin x \cosh y + i \cos x \sinh y$$

$$= r e^{i\theta}$$

$$r = \sqrt{\sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y}$$

$$= \sqrt{(1 - \cos^2 x) \cosh^2 y + \cos^2 x (\cosh^2 y - 1)}$$

$$= \sqrt{\cosh^2 y - \cancel{\cos^2 x} \cosh^2 y + \cancel{\cos^2 x} \cosh^2 y - \cos^2 x}$$

$$= \sqrt{\cosh^2 y - \cos^2 x}$$

$$\theta = \tan^{-1} \left(\frac{\cos x \sinh y}{\sin x \cosh y} \right) = \tan^{-1} (\cot x \tanh y)$$

$$\log [\sin(x + iy)] = \log (r e^{i\theta}) = \log r + i\theta$$

$$= \log r + i\theta$$

$$= \frac{1}{2} \log (\cosh^2 y - \cos^2 x) + i \tan^{-1} (\cot x \tanh y)$$

- Prove that $\log(1 + e^{2i\theta}) = \log(2 \cos \theta) + i\theta$

$$\begin{aligned} & \log(1 + \cos 2\theta + i \sin 2\theta) \\ &= \log(2 \cos^2 \theta + i 2 \sin \theta \cos \theta) \\ &= \log(2 \cos \theta (\cos \theta + i \sin \theta)) \\ &= \log(\underbrace{2 \cos \theta}_{\text{}} e^{i\theta}) \\ &= \log(2 \cos \theta) + i\theta \end{aligned}$$

- Separate into real and imaginary parts $\sqrt{i}^{\sqrt{i}}$

$$z = (\sqrt{i})^{\sqrt{i}}$$

$$\log z = \sqrt{i} \log \sqrt{i}$$

$$= \sqrt{i} \frac{1}{2} \log i$$

$$= \sqrt{i} \frac{1}{2} \log (e^{i\pi/2})$$

$$= \sqrt{i} \frac{1}{2} \frac{\pi}{2} = \frac{i\pi}{4} \sqrt{i}$$

$$= \frac{i\pi}{4} (i)^{1/2} = \frac{i\pi}{4} (e^{i\pi/2})^{1/2}$$

$$= \frac{i\pi}{4} e^{i\pi/4}$$

$$= \frac{i\pi}{4} (\cos \pi/4 + i \sin \pi/4)$$

$$= i\pi/4 \left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \right)$$

$$\log z = \frac{i\pi}{4\sqrt{2}} - \frac{\pi}{4\sqrt{2}}$$

$$z = e^{i\pi/4\sqrt{2} - \pi/4\sqrt{2}}$$

$$= e^{-\pi/4\sqrt{2}} e^{i\pi/4\sqrt{2}}$$

$$= e^{-\pi/4\sqrt{2}} \left(\cos \frac{\pi}{4\sqrt{2}} + i \sin \frac{\pi}{4\sqrt{2}} \right)$$

- Find the principal value of $(1+i)^{1-i}$

$$z = (1+i)^{1-i}$$

$$\log z = \log (1+i)^{1-i}$$

$$= (1-i) \log(1+i)$$

$$= (1-i) \log(\sqrt{2} e^{i\pi/4})$$

$$= (1-i) \left(\log \sqrt{2} + i\pi/4 \right)$$

- Find the principal value of $(1+i)^{1-i}$

$$z = (1+i)^{1-i}$$

$$\log z = (1-i) \log(1+i)$$

$$= (1-i) \log(\sqrt{2} e^{i\pi/4})$$

$$= (1-i) (\log \sqrt{2} + i\pi/4)$$

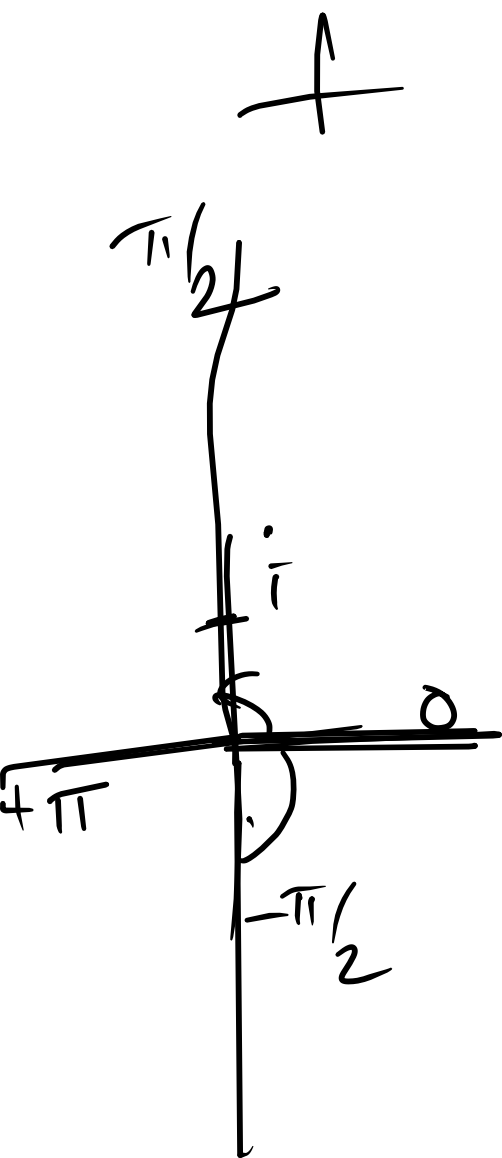
$$= \log \sqrt{2} - i \log \sqrt{2} + i\pi/4 + \pi/4$$

$$\log z = (\log \sqrt{2} + \pi/4) + i(\pi/4 - \log \sqrt{2})$$

$$z = e^{(\log \sqrt{2} + \pi/4) + i(\pi/4 - \log \sqrt{2})}$$

$$= e^{(\log \sqrt{2} + \pi/4)} e^{i(\pi/4 - \log \sqrt{2})}$$

$$= \sqrt{2} e^{\pi/4} \left(\cos(\pi/4 - \log \sqrt{2}) + i \sin(\pi/4 - \log \sqrt{2}) \right)$$



- Prove that the general value of $(1 + i \tan \alpha)^{-i}$ is
- $e^{2m\pi + \alpha} [\cos(\log \cos \alpha) + i \sin(\log \cos \alpha)]$

$$\begin{aligned}
 z &= (1 + i \tan \alpha)^{-i} \\
 \log z &= -i \log (1 + i \tan \alpha) \\
 &= -i \log (r e^{i\theta}) \\
 &= -i \log (r e^{i(2m\pi + \theta)}) \\
 &= -i [\log r + i(2m\pi + \theta)]
 \end{aligned}$$

$$\text{when, } r = \sqrt{1 + \tan^2 \alpha} = \sec \alpha$$

$$\theta = \tan^{-1}(\tan \alpha) = \alpha$$

$$\begin{aligned}
 \log z &= -i [\log(\sec \alpha) + i(2m\pi + \alpha)] \\
 &= (2m\pi + \alpha) - i(\log(\sec \alpha))
 \end{aligned}$$

$$\begin{aligned}
 z &= e^{(2m\pi + \alpha) - i(\log(\sec \alpha))} \\
 &= e^{(2m\pi + \alpha)} e^{-i(\log(\sec \alpha))} = e^{(2m\pi + \alpha)} e^{i(\log(\sec \alpha))} = e^{(2m\pi + \alpha)} e^{i \log(\cos \alpha)} \\
 &= e^{(2m\pi + \alpha)} [\cos(\log(\cos \alpha)) + i \sin(\log(\cos \alpha))]
 \end{aligned}$$

- If $\frac{(a+ib)^{x+iy}}{(a-ib)^{x-iy}} = \alpha + i\beta$, find α and β

$$\log \left(\frac{(a+ib)^{x+iy}}{(a-ib)^{x-iy}} \right) = \log(\alpha + i\beta)$$

$$\log(a+ib)^{x+iy} - \log(a-ib)^{x-iy} = \log(\alpha + i\beta)$$

$$\begin{aligned} \log(\alpha + i\beta) &= (x+iy) \log(a+ib) - (x-iy) \log(a-ib) \\ &= (x+iy) \log(re^{i\theta}) - (x-iy) \log(re^{-i\theta}) \end{aligned}$$

$$r = \sqrt{a^2 + b^2} \quad \theta = \tan^{-1}\left(\frac{b}{a}\right)$$

$$= (x+iy)(\log r + i\theta) - (x-iy)(\log r - i\theta)$$

$$= \cancel{x \log r} + i \cancel{x \theta} + iy \log r - \cancel{y \theta} - (\cancel{x \log r} - i \cancel{x \theta} - iy \log r - \cancel{y \theta})$$

$$= 2iy \log r + 2i x \theta$$

$$\log(\alpha + i\beta) = i 2 (x \theta + y \log r)$$

$$= i 2 \left(x \tan^{-1}\left(\frac{b}{a}\right) + \frac{y}{2} \log(a^2 + b^2) \right)$$

(consider

$$m = x \tan^{-1}\left(\frac{b}{a}\right) + \frac{y}{2} \log(a^2 + b^2)$$

$$\log(\alpha + i\beta) = i 2m$$

$$(\alpha + i\beta) = e^{i 2m} = \cos(2m) + i \sin(2m)$$

where $m =$

• If $i^{\alpha+i\beta} = \alpha + i\beta$ (or $i^{i^{\infty}} = \alpha + i\beta$), prove that $\alpha^2 + \beta^2 = e^{-(4n+1)\pi\beta}$ Where n is any positive integer

$$i^{i^{\infty}} = \alpha + i\beta$$

$$i^{(\alpha+i\beta)} = \alpha + i\beta$$

$$\log(\alpha + i\beta) = \log i^{(\alpha+i\beta)}$$

$$= (\alpha + i\beta) \log(e^{i\pi/2})$$

$$= (\alpha + i\beta) \log(e^{i(2n\pi + \pi/2)})$$

$$= (\alpha + i\beta) i(2n\pi + \pi/2)$$

$$\log(\alpha + i\beta) = (\alpha + i\beta) \frac{i}{2}(4n\pi + \pi)$$

$$= \left(i2n\pi + \frac{i\pi}{2}\right)(\alpha + i\beta)$$

$$= \alpha i2n\pi + \alpha \frac{i\pi}{2} - 2\beta n\pi - \frac{\beta\pi}{2}$$

$$\log(\alpha + i\beta) = \left(-2\beta n\pi - \frac{\beta\pi}{2}\right) + i\left(2n\pi\alpha + \frac{\alpha\pi}{2}\right)$$

$$\alpha + i\beta = e^{-2\beta n\pi - \frac{\beta\pi}{2}} e^{i(2n\pi\alpha + \frac{\alpha\pi}{2})}$$

$$= r e^{i\theta}$$

$$r = e^{-2\beta n\pi - \frac{\beta\pi}{2}}$$

$$\theta = 2n\pi\alpha + \frac{\alpha\pi}{2}$$

$$= r(\cos\theta + i\sin\theta)$$

$$\alpha = r\cos\theta, \quad \beta = r\sin\theta$$

$$\alpha^2 + \beta^2 = r^2 = \left(e^{-2\beta n\pi - \frac{\beta\pi}{2}}\right)^2$$

$$= \left(e^{-4\beta n\pi - \beta\pi}\right)$$

$$= e^{-\beta\pi(4n+1)}$$